Debojjal Bagchi

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Contact Information

PhD Candidate & Graduate Research Assistant
Fariborz Maseeh Department of Civil, Architectural, and Environmental Engineering
The University of Texas at Austin
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Research Interests

Freight and maritime logistics; traffic assignment and network modeling; resilience assessment.

Education

The University of Texas at Austin, Austin, TX

Doctor of Philosophy — 2023 – 2028
Major: Civil Engineering (Transportation Systems)
GPA: 4.00 / 4.00
Committee: Dr. Stephen D. Boyles (Supervisor), Dr. Chandra R. Bhat, Dr. Hillel Bar-Gera, Dr. Raghu Bollapragada, Dr. Zhaomiao Guo

Master of Science in Engineering (Thesis) — 2023 – 2025
Major: Civil Engineering (Transportation Systems)
GPA: 4.00 / 4.00
Thesis: Error bounds for stochastic user equilibrium traffic assignment [Link]
Award: **CUTC Milton Pikarsky Memorial Award for outstanding MS thesis in science and technology**
Advisor: Dr. Stephen D. Boyles; *Reader:* Dr. Zhaomiao Guo

Indian Institute of Science (IISc), Bengaluru, India

Bachelor of Science (Research) — 2019 – 2023
Major: Earth & Environmental Science, *Minor:* Mathematics
GPA: 8.7 / 10.0; *Major GPA:* 9.4 / 10.0
Thesis: Energy-efficient and safe routing for last-mile logistics [Link]
Award: **Institute Gold Medal for highest major GPA**
Advisor: Dr. Tarun Rambha

Professional Service

Standing Committee AT014, Transportation Research Board — 2025 – 2028
Committee Member, Intermodal Freight and Truck Transportation [Link]

Standing Committee AW010, Transportation Research Board — 2025
Committee Member, Ports and Channels

Awards and Fellowships

International & National Awards

Milton Pikarsky Memorial Award — 2025
Outstanding master's thesis in science and technology, Council of University Transportation Centers, USA.

MITACS Globalink Research Internship (GRI) Award — 2022
Fully funded summer research at a Canadian university, MITACS, Canada.

Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship — 2019
Fully funded undergraduate scholarship and research grant, Department of Science and Technology, Government of India.

Dhirubhai Ambani Scholarship — 2019
Awarded to top-ranking students in India's Class 12 board examinations, Reliance Foundation, India.

National Talent Search Examination (NTSE) Scholarship — 2017
High school and pre-college scholarship awarded to 1,000 students across India, National Council of Educational Research and Training, Government of India.

Regional & University-Level Awards

TexITE Student Paper Award 2026
Second prize, TexITE.

Best Poster Award 2026
Graduate Symposium, Department of Civil, Architectural, and Environmental Engineering, UT Austin.

Professional Development Award 2025
Graduate School, UT Austin.

UT Austin Engineering Fellowship 2023 – 2027
Cockrell School of Engineering and Graduate School, UT Austin.

Institute Gold Medal 2023
Indian Institute of Science, awarded for highest major GPA in undergraduate studies.

Telegraph School Award 2019
Special honor for academic excellence by a student.

Jagadish Bose National Science Talent Search (JBNSTS) Fellowship 2018

Sponsored Research Experience

Graduate Research Assistantship Jan 2026 – Present
"Develop Multi-Modal Maritime-Rail-Roadway Transportation Model for the Texas Inland and Intercoastal Waterways," sponsored by the Texas Department of Transportation
PIs: Dr. Stephen D. Boyles and Dr. C. Tyler Dick
- Assisted with proposal authorship.
- Developed a statewide multimodal freight model for Texas.
- Collected freight commodity, volume, and routing data for road, rail, and maritime modes.
- Conducted risk assessments to quantify disaster impacts on the Texas freight network.
- Tested scenarios to identify travel time variations and critical links.

Graduate Research Assistantship Sep 2025 – Present
"Evaluating Texas Ports Readiness and Opportunities for Alternative Fuels," sponsored by the Texas Department of Transportation
PIs: Dr. Andrea Gold and Dr. Anna McAuley
- Investigated the readiness of Texas ports for alternative fuels.
- Conducted stakeholder outreach to collect data on system fuel capabilities.
- Identified gaps and opportunities in port alternative fuel systems.
- Performed market demand analysis for alternate fuel use at Texas ports.
- Developed a tool to assess port readiness for alternative fuels.

Graduate Research Assistantship Sep 2023 – Dec 2024
"Data-Driven Multimodal Freight Modeling for Waterways and Ports," sponsored by Coastal and Hydraulics Laboratory, U.S. Army Engineer Research and Development Center
PI: Dr. Stephen D. Boyles
- Integrated multimodal data sources, including Automatic Identification System vessel data.
- Built a discrete-event simulation of the Port of Houston waterway, terminals, and landside.
- Developed queueing and differential equation models for capacity and bottleneck analysis.
- Simulated hurricane and fog disruptions to assess port performance with capacity limits.
- **Led to 5 conference talks and 4 journal articles (2 published, 2 in preparation).**

MITACS Globalink Research Internship May 2022 – Aug 2022
"Integrating Waste and Resource Management: Data-Driven Optimization of Urban Mining Logistics," sponsored by MITACS at Université du Québec à Trois-Rivières
PI: Dr. Amina Lamghari
- Studied reverse logistics network design under uncertainty for wood industries in Quebec.
- Developed a scenario-based Mixed Integer Linear Program (MILP) for network design.
- Developed an Adaptive Large Neighborhood Search (ALNS) heuristic for faster solving.
- Solved the MILP using CPLEX and compared it against the ALNS heuristic in Python.

Undergraduate Research Internship Jul 2021 – Jul 2023

"A Local Search Heuristic for the Bi-criterion Steiner Traveling Salesperson Problem with Time Windows (BSTSPTW)," at the Indian Institute of Science, Bengaluru, India
PI: Dr. Tarun Rambha

- Extended the Lin-Kernighan-Helsgaun (LKH) heuristic to BSTSPTW.
- Formulated a scalarization-based, single-commodity flow MILP for BSTSPTW.
- Proposed a heuristic that beat LKH and the MILP on Amazon routes in Austin, TX.
- **Resulted in two conference presentations and a journal article.**

Peer-Reviewed Publications

[P3] Bathgate, K., **Bagchi, D.**, Mitchell, K. N., Asborno, M. I., Kress, M. M., and Boyles, S. D. (2026). Anchorage Queue Dynamics due to Fog Channel Closures at the Port of Houston. *Journal of Waterway, Port, Coastal, and Ocean Engineering.* DOI: 10.1061/jwped5.wweng-2451

[P2] Bathgate, K., **Bagchi, D.**, Mitchell, K. N., Asborno, M. I., Kress, M. M., and Boyles, S. D. (2026). Characterizing Longitudinal Vessel Arrival and Queue Behavior from AIS at the Port of Houston. *Journal of Waterway, Port, Coastal, and Ocean Engineering.* DOI: 10.1061/jwped5.wweng-2441

[P1] Agarwal, P.†, **Bagchi, D.**†, Rambha, T., and Pandey, V. (2025). A Bi-criterion Steiner Traveling Salesperson Problem with Time Windows for Last-Mile Electric Vehicle Logistics. *Computers & Operations Research.* DOI: 10.1016/j.cor.2025.107286 [Code] (†**Equal contribution**)

Manuscripts Under Review and Preprints

[R5] **Bagchi, D.** and Boyles, S. D. (2026). Error Bounds for Stochastic User Equilibrium Traffic Assignment. *Under review at Transportation Science.*

[R4] **Bagchi, D.**†, Bathgate, K.†, Mitchell, K. N., Asborno, M. I., Kress, M. M., and Boyles, S. D. (2026). A Simulation Framework for Evaluating Impacts of Expansion Projects on Multimodal Port System Capacity and Resilience. *Under review at Simulation Modelling Practice and Theory.* DOI: 10.2139/ssrn.7447899 [Code] (†**Equal contribution**)

[R3] **Bagchi, D.** and Boyles, S. D. (2026). Spectral Analysis of the Logit Mapping and Implications for Stochastic User Equilibrium Algorithms. *Under review at Transportation Research Part B: Methodological.* arXiv:2605.21843 [Code]

[R2] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2026). Dynamic Traffic Assignment with Uncertain Demand: A Localized Queue Spillback Approach. *Under review at Transportmetrica B: Transport Dynamics.*

[R1] **Bagchi, D.**, Bathgate, K., Mitchell, K. N., Asborno, M. I., Kress, M. M., and Boyles, S. D. (2025). Measuring Capacities in Multimodal Maritime Port Systems with Anchorage Queues. *Preprint.* arXiv:2509.22961

Peer-Reviewed Conference Presentations

[C11] **Bagchi, D.** and Boyles, S. D. (2027). Spectral Analysis of the Logit Mapping and Implications for Stochastic User Equilibrium Algorithms. *Transportation Research Board 106th Annual Meeting* (**TRB**), Washington, D.C., USA (Lectern session).

[C10] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2027). Resilience Metrics for Tropical Cyclone Anchorage Queue Responses. *Transportation Research Board 106th Annual Meeting* (**TRB**), Washington, D.C., USA (Poster session).

[C9] **Bagchi, D.**, Bathgate, K., Boyles, S. D., Mitchell, K. N., Asborno, M. I., and Kress, M. M. (2027). A Simulation Framework for Evaluating Impacts of Expansion Projects on Multimodal Port System Capacity and Resilience. *Transportation Research Board 106th Annual Meeting* (**TRB**), Washington, D.C., USA (Poster session).

[C8] **Bagchi, D.**, Bathgate, K., Boyles, S. D., and Asborno, M. I. (2026). A Dynamic Ultimate Capacity Framework for Seaport Systems. *Transportation Research Board 105th Annual Meeting* (**TRB**), Washington, D.C., USA (Lectern session).

[C7] Bathgate, K., **Bagchi, D.**, Boyles, S. D., Mitchell, K. N., Asborno, M. I., and Kress, M. M. (2026). Characterizing Anchorage Queue Dynamics due to Channel Fog Closures at the Port of Houston. *Transportation Research Board 105th Annual Meeting* (**TRB**), Washington, D.C., USA (Poster session).

[C6] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2026). Dynamic Traffic Assignment with Uncertain Demand: A Localized Queue Spillback Approach. *Transportation Research Board 105th Annual Meeting* (**TRB**), Washington, D.C., USA (Poster session).

[C5] Robbennolt, J., **Bagchi, D.**, and Boyles, S. D. (2025). Localized Queue Spillback with Uncertain Demand. *10th International Symposium on Dynamic Traffic Assignment* (**DTA2025**), Salerno, Italy (Lectern session).

[C4] **Bagchi, D.** and Boyles, S. D. (2025). Error Bounds for Stochastic User Equilibrium Traffic Assignment. *12th Triennial Symposium on Transportation Analysis* (**TRISTAN XII**), Okinawa, Japan (Lectern session).

[C3] **Bagchi, D.**, Bathgate, K., and Boyles, S. D. (2025). A Queuing-theory-based Operating Capacity Model for Multimodal Port Operations. *Transportation Research Board 104th Annual Meeting* (**TRB**), Washington, D.C., USA (Lectern session).

[C2] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2025). Use of AIS Data to Characterize Vessel Mix in Houston Port Operations for Simulation. *Transportation Research Board 104th Annual Meeting* (**TRB**), Washington, D.C., USA (Lectern session).

[C1] **Bagchi, D.**, Agarwal, P., Rambha, T., and Pandey, V. (2023). A Local Search Heuristic for Bi-criterion Steiner Traveling Salesperson Problem. *Transportation Research Board 102nd Annual Meeting* (**TRB**), Washington, D.C., USA (Poster session).

Invited Presentations

[I4] **Bagchi, D.** and Boyles, S. D. (2026). Theoretical and Algorithmic Advances in Logit-Based Stochastic User Equilibrium Traffic Assignment. *Institute for Operations Research and the Management Sciences Annual Meeting 2026* (**INFORMS**), San Francisco, USA (Invited session).

[I3] **Bagchi, D.**, Bathgate, K., and Boyles, S. D. (2025). A Framework for Measuring Maritime Port System Capacities with Limited Input Data. *Institute for Operations Research and the Management Sciences Annual Meeting 2025* (**INFORMS**), Atlanta, USA (Invited session).

[I2] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2024). Identifying Capacities in a Multimodal Maritime Freight Network. *Institute for Operations Research and the Management Sciences Annual Meeting 2024* (**INFORMS**), Seattle, USA (Invited session).

[I1] **Bagchi, D.** and Boyles, S. D. (2024). Error Bounds for Stochastic User Equilibrium Traffic Assignment. *Institute for Operations Research and the Management Sciences Annual Meeting 2024* (**INFORMS**), Seattle, USA (Invited session).

Technical Presentations

[T3] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2025). Simulation Methods for Measuring Multimodal Maritime Freight System Capacity and Resilience: A Case Study at the Port of Houston. *UT Austin Center for Transportation Research Annual Symposium, 2025*, Austin, USA.

[T2] Bathgate, K., **Bagchi, D.**, and Boyles, S. D. (2024). Data-Driven Modeling for Multimodal Port Resilience Assessment. *UT Austin Center for Transportation Research Annual Symposium, 2024*, Austin, USA.

[T1] **Bagchi, D.**, Agarwal, P., Rambha, T., and Pandey, V. (2022). A Local Search Heuristic for Bi-criterion Steiner Traveling Salesperson Problem. *Institute for Operations Research and the Management Sciences Annual Meeting 2022* (**INFORMS**), Indianapolis, USA (Contributed session)..

Open Source Projects

Logit-SUE [Code]
Developer
- Among the fastest open-source Python solvers (as of 2026) for path-based logit SUE.
- Implements Newton-GMRES with superlinear convergence and Barzilai-Borwein rules.
- Includes Method of Successive Averages variants with adaptive constant step sizes.
- Benchmarked on large-scale test networks, including Chicago Regional and Philadelphia.

Multimodal Analysis Port Simulation (MAPS) [Code] [Technical manual]
Lead developer
- Modular Python discrete-event simulation of container, liquid, and dry bulk terminals.
- Covers waterway and landside operations, including rail and roadways.
- Supports planning, bottleneck detection, and what-if scenario analysis.
- Uses real-world data inputs; sample port data included.
- Models capacity and resilience under channel expansion, hurricane, and fog scenarios.

Last-Mile EV Logistics [Code]
Co-developer
- Safe and energy-efficient routing of last-mile electric freight vehicles.
- Tested on Amazon delivery routes in Austin, TX (data included).
- Solves the bi-criterion Steiner TSP with turn penalties using state-of-the-art heuristics.
- Also includes exact approaches via a Mixed Integer Linear Program solved with CPLEX.

Teaching Experience

Teaching Assistant, University of Texas at Austin
CE 311S Probability and Statistics for Civil Engineers — Fall 2025
Consolidated student feedback rating: 4.22 / 5.00
- Led one weekly discussion and problem-solving section of 20 students for one hour.
- Conducted one weekly one-hour office hour session.
- Prepared assignment solutions and assisted with grading.

CE 311S Probability and Statistics for Civil Engineers — Spring 2025
Consolidated student feedback rating: 4.02 / 5.00
- Assisted with a major course redesign.
- Led two weekly one-hour discussion and problem-solving sections of 20 students each.
- Conducted two weekly one-hour office hour sessions.
- Prepared assignment solutions and assisted with grading.

Engineering Teaching Assistant Certificate — 2025
Cockrell School of Engineering, University of Texas at Austin.

Peer Review and Society Service

Journal Referee
- Transportation Research Part B (2 reviews).
- Transportation Research Record (1 review).

Conference Referee
- Transportation Research Board Annual Meeting (17 reviews).

Society Service

Community Service Chair, UT Student Transportation Council — 2023 – 2025
UT Austin chapter of the Institute of Transportation Engineers

Member, Notebook Drive, Indian Institute of Science — 2019 – 2023